

MD. MAHDI HASAN

 Md. Mahdi Hasan |  mahdi.hasan01@northsouth.edu , mahdi.hasan2212039@gmail.com |  +880 1521 725630

Research Interests

Deep Learning and Computer Vision, Visual Representation Learning, Efficient and Multi-Modal AI, Attention-Based Deep Learning, Object Detection and Image Understanding, Image Forensics, Explainable and Robust AI, Resource-Efficient and Edge AI, and Real-Time Intelligent Systems for UAV-Based Precision Agriculture.

Education

North South University

Bachelor of Science in Computer Science and Engineering

Dhaka, Bangladesh

January 2022 – May 2026

- **Summa Cum Laude Distinction** (CGPA: 3.80 / 4.00)
- **Merit-Based Scholarship:** 75% Tuition Scholarship
- **Relevant Coursework:** Machine Learning, Pattern Recognition, Theory of Computation, and Introduction to Robotics

Research Experience

Undergraduate Researcher

Electrical and Computer Engineering, North South University

May 2025 – Present

Dhaka, Bangladesh

- Developed an edge-ready UAV-based deep learning system for real-time simultaneous detection of weeds and pests using attention-enhanced architectures, including MoonNet and dual-head CerberusDet, achieving up to 91% mAP for insect detection and demonstrating deployment on Raspberry Pi 4 for precision agriculture.
- Designed HyDAE-Net, an efficient multimodal deep learning framework for localized image forgery detection by fusing RGB ConvNeXtV2 semantic features with ConvFormer noise-artifact features through deformable cross-modal attention. Achieved 0.83 IoU and 0.87 F1-score on the BR-Gen dataset while reducing model size by 35% compared with state-of-the-art models.
- Conducted research in deep learning and computer vision, focusing on efficient neural architectures, attention mechanisms, object detection, multimodal feature fusion, and edge deployment. Two research manuscripts are currently under review.

Teaching Experience

Lab Instructor — Dept. of Electrical & Computer Engineering

North South University

June 2026 – Present

Dhaka, Bangladesh

- Conduct **9 hours of laboratory classes weekly** across **CSE 332L — Computer Organization & Design** and **CSE 231L — Digital Logic Design**, delivering theoretical instruction and hands-on technical guidance to students.
- Hold **11 hours of weekly office hours** to support students with laboratory concepts, hardware/software implementation, debugging, and project development.
- Design and prepare laboratory assessments, quizzes, practical examinations, assignments, and evaluation rubrics aligned with course learning objectives.
- Grade laboratory assignments, quizzes, practical examinations, and project work, providing constructive feedback and maintaining accurate assessment records.
- Prepare and organize laboratory materials, demonstrate experimental procedures, troubleshoot student implementations, and ensure effective laboratory class conduction.

Teaching Assistant — Dept. of Electrical & Computer Engineering

North South University

January 2025 – May 2026

Dhaka, Bangladesh

- Supported **350+ students** across three courses: **Computer Organization & Architecture (CSE332)**, **Operating Systems Design (CSE323)**, and **Microprocessor Interfacing & Embedded Systems (CSE331)**.
- Led weekly problem-solving sessions and office hours, providing conceptual clarification, exam preparation, and targeted academic guidance.
- Graded assignments and examinations contributing to **10% of the total course grade**, provided constructive feedback, and assisted faculty with proctoring and classroom management.

Teaching Assistant — Dept. of Mathematics & Physics

North South University

January 2024 – December 2024

Dhaka, Bangladesh

- Supported **250+ students** across two courses: **MAT350 (Engineering Mathematics)** and **MAT250 (Calculus II)**.
- Led weekly problem-solving sessions, conducted office hours for conceptual clarification and exam preparation, and provided targeted academic guidance.
- Graded assignments and examinations contributing to **30% of the total course grade**, provided constructive feedback, and assisted faculty with proctoring and classroom management.

Industry Experience

Machine Learning Intern

July 2026 – Present

FlyRank AI

Remote

- Applied **Python, pandas, scikit-learn, and sentence-transformers** to analyze real-world search and analytics datasets and develop data-driven insights.
- Performed **data wrangling, exploratory analysis, semantic analysis, and text clustering** on complex datasets to identify search intent and content gaps.
- Developed **reproducible Jupyter/Google Colab notebooks** and translated ML findings into actionable optimization recommendations for real-world business problems.

Ongoing Papers

- M. M. Hasan, A. Rahman, T. Ferdoush, M. Hossain, and R. Khan, “Real-Time UAV-Based Multi-Task Detection of Weeds and Pests Using a Multi-Head YOLO Architecture on Edge Devices,” *[Ongoing Manuscript]*.
- M. Sadat, M. Rahman, M. M. Hasan, M. M. I. Khan, and R. Khan, “HyDAE-Net: A Hybrid Dual-Stream Deformable Attention Framework for Efficient Localized Image Forgery Detection,” *[Under Review]*.
- M. M. Hasan, M. I. A. Rafid, and S. Siddique, “ARES: An Energy-Aware Simulation Framework for Autonomous Planetary Navigation & Perception,” *[Manuscript in Preparation for Conference Submission]*.

Skills

- **Programming Languages:** Python, C, C++, Java, HTML, CSS
- **Data Science & Machine Learning:** TensorFlow, PyTorch, Keras, HuggingFace, Scikit-learn
- **Developer Tools:** Git, MySQL, Pandas, NumPy, Matplotlib, Seaborn, Django, Flask, Kaggle, Android Studio, Jupyter, Trello, VSCode
- **Documentation & Presentation Tools:** LaTeX, Google Workspace, Microsoft Office Suite

References

Dr. Riasat Khan

Associate Professor

Department of Electrical & Computer Engineering

North South University, Dhaka, Bangladesh

Email: riasat.khan@northsouth.edu

Phone: +88 02 55668200 Ext. 6382

Dr. Shahnewaz Siddique

Associate Professor

Department of Electrical & Computer Engineering

North South University, Dhaka, Bangladesh

Email: shahnewaz.siddique@northsouth.edu

Phone: +88 02 55668200 Ext. 1515

Dr. Preetom Nag

Associate Professor

Department of Mathematics & Physics

North South University, Dhaka, Bangladesh

Email: preetom.nag@northsouth.edu

Phone: +88 02 55668200 Ext. 6219